

36 A bacterial pathogen produces a protein that acts as a toxin. This toxin is harmful to humans.

Scientists are developing monoclonal antibodies that can be used to detect the presence of the toxin in the body so that early treatment can be given.

Which statements describe steps in the development of these monoclonal antibodies?

- 1 The toxin protein is injected into a mouse and triggers mitosis of specific B-lymphocytes.
- 2 Antibodies are collected from the spleen of the mouse and fused with myeloma cells.
- 3 A hybridoma cell produces many antibodies with a variety of different variable regions.

A 1 and 2

B 1 only

C 2 and 3

D 3 only